Sem 5 2023-24 CLOUD MACHINE LEARNING

Section 1 : Case study - 40 marks.

NSS wants to develop AI powered chatbots to handle customer inquiries provide support and process common service requests.

Goal: Reduce response time and improve customer satisfaction

Q1 Prepare a Comprehensive report.

Q1(a). Identify and evaluate the current cloud computing technologies available for implementing an AI/ML chatbots. Discuss strengths and weakness to justify your choice of cloud AI/ML services and tools to implement one. (15 Marks)

Ans: Evaluation of cloud Computing Technologies for AI/ML chatbot ie to implement an AI powered chatbot, NSS may use cloud platform such as:

a) AWS
b) Azure.
c) Google cloud platform.

AWS	Azure	GCP
Key services- Amazon Lex (Chatbot builder)	Azure bot service	Dialogflow.
AWS Lambda.	Azure cognitive servic.	Vertex AI.
Amazon Sage Maker.	Azure machine Learning	
Strengths → Highly scalable & reliable.	→ Easy integration with enterprise tools (Eg- office, teams).	→ Best in class NLP (Dialogflow)
→ Strong NLP (Natural Language processing) with Lex.	→ Strong NLP via cognitive services	→ Easy chatbot training and development
→ Large ecosystem and integrations	→ Developer friendly	→ Strong AI/ML research backing..
Weakness → Complex pricing model	→ cost can increase with scaling	→ Smaller enterprise ecosystem than Azure
→ Steep learning curve for beginners	→ Slightly less flexible than AWS in customization	→ Limited regional availability compared to AWS.

Justification of choice.

For NSS GCP is the most suitable choice because.

→ Dialogflow simplifies chatbot development
→ Superior Natural Language understanding (NLU)
→ Fastest development cycle → ideal for client facing solution.
→ Lower complexity for mid sized teams.

Q1(b). How would you design and deploy an AI powered chatbot for the services and tools chosen in the above part? (10 Marks)

~~Architecture Design~~ ~~Development Steps~~

Architecture Design

Components:
UI (web / Mobile / whatsapp)
Dialogflow (NLP engine)
Backend (API via cloud fun)
Database (Firestore / cloud SQL)

Development Steps

Step 1: Requirement Analysis
→ Identify user queries (FAQs, service requests)
Step 2: Intent & Entity Design.
Eg:. Create order status
Raise complaint
Step 3: Model training
→ Train chatbot using sample phrases.
→ Use Dialogflow ML to improve accuracy.
Step 4: Backend integration
→ Connect chatbot to APIs using cloud functions
Step 5: Testing.
→ perform unit & user testing.
→ Improve based on feedback.

Deployment

Deploy chatbot on:
- website
→ Mobile app.
→ Messaging platform (whatsapp, Teams)
→ Use CI/CD pipeline. for updates.

Q1(c). What metrics would you use to measure the success of the chatbot?

Ans: Accuracy Rate → percentage of correctly understood user queries. (5 Marks)
Response Time → Time takes to respond user input
Customer Satisfaction → user feedback after interaction
Resolution Rate → percentage of queries resolved without human intervention
Fallback Rate → No: of queries the chatbot fails to understand.

Additionally a confusion matrix can be used formally to evaluate the performance of the ~~cat~~ chatbot's intent classification model by analysing true positives, false positives and false negatives.

Q1(d) Discuss the ethical implications of using these technologies, particularly in terms of data, privacy, security and bias. Recommend potential risks and mitigation strategies

Ans: **Data privacy**

Risk: Exposure of personal customer data.

Mitigation: Use encryption (TLS, AES), Follow GDPR
Minimize data collection

Q.1 (d) Contradictions Security

Risk → unauthorized Access / Cyberattacks

Mitigation: → Multi-factor authentication.
Role based access control (RBAC)
Regular security audits

Bias in AI models Risk: chatbot may give biased or unfair responses.

Mitigation: Use diverse training datasets
Regular bias ~~testing~~ testing.
Human review loop.

Transparency Risk: users unaware they are interacting with AI

Mitigation: → clearly disclose chatbot identity.
→ provide option to talk to human agent.

Accountability Risk: Incorrect decisions affecting users.

Mitigation: → Logging and monitoring.
→ Escalation mechanisms.
→ Human in the loop systems.

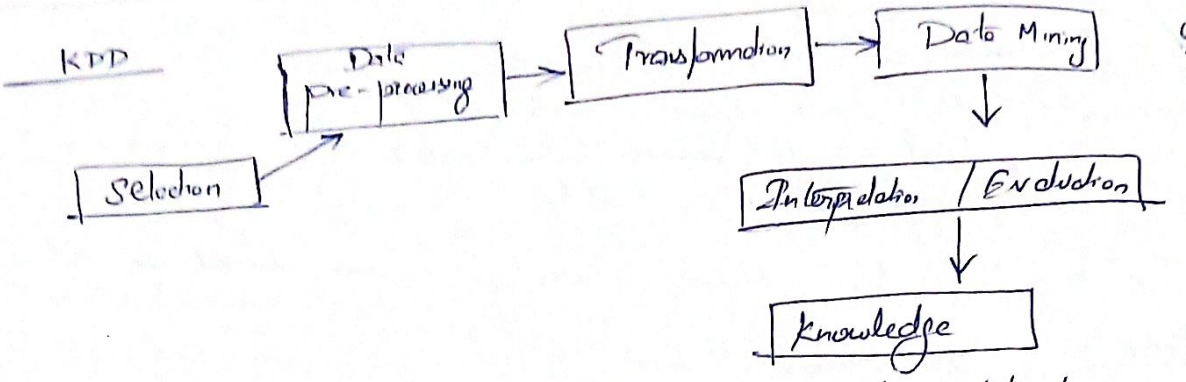
Final conclusion By adopting CICP dialogflow based architecture.
NJ can build a scalable, intelligent chatbot that enhances customer
experience with maintaining security, ethical compliance and high performance.

Q.2 (a) Compare and Contrast the data science methodologies: KDD, CRISP-DM and SEMMA

Data science methodologies provide structured approaches for extracting knowledge and insights from data.

Three widely used methodologies.

* KDD (Knowledge discovery in Databases)
* CRISP-DM (Cross Industrial Standard procedure for Data Mining)
* SEMMA (Sample, Explore, Modify, Model and Asses)



ie. ① Selection — Select the data from large dataset
② pre-processing — clean and preprocess the data
③ Transformation — Transform the data into need for analysis
④ Data Mining — Apply algorithms & patterns to the data.
⑤ Interpretation / Evaluation — Evaluating the data pattern
⑥ Knowledge → Getting the results.

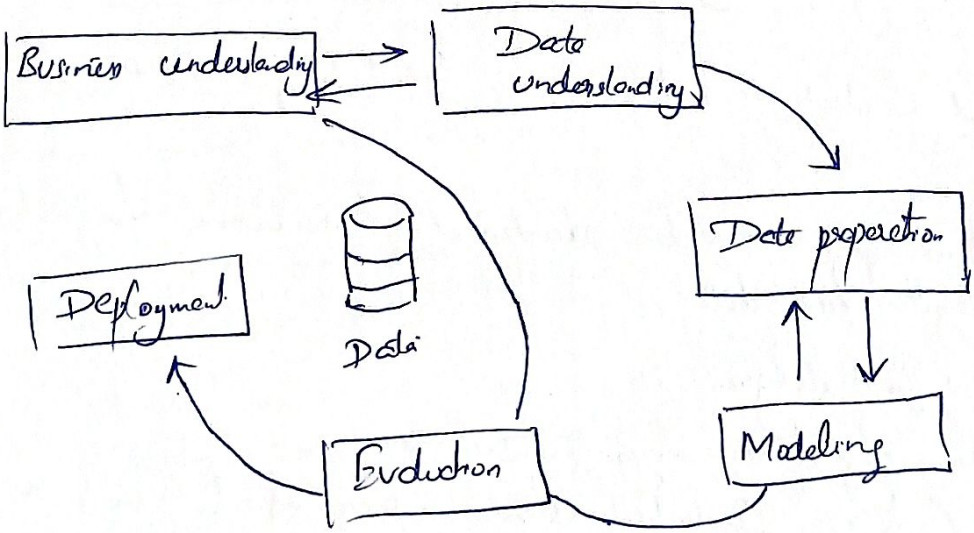
Advantages / strengths

* Strong focus on knowledge extraction
* Good for research oriented projects

weakness

→ Less business oriented
→ Does not clearly address deployment & business understanding.

CRISP-DM (Cross Industrial Standard procedure for Data Mining)



ie ① Business understanding
② Data understanding
③ Data preparation
④ Modeling
⑤ Evaluation
⑥ Deployment

Advantages

→ Industry standard
→ Provide a structured approach to data mining.

Disadvantages

→ Can be rigid
→ time consuming.

SEMMA → developed by SAS (Statistical Analysis System)

Steps in SEMMA
1. Sample
2. Explore
3. Modify
4. Model
5. Assess – Evaluating the data

Advantages
- Strong focus on modeling & analysis
- Efficient for large datasets

Disadvantages
- Less emphasis on business understanding
- Highly tool oriented towards SAS environment

Conclusion: Among 3 methodologies, CRISP-DM is the most widely adopted because it provides a complete business oriented framework from problem understanding to deployment. KDD is more research focused SEMMA model centric and SAS analysis tool.

2(b). 3 characteristics of ML algorithms
1. Unsupervised learning
2. Supervised learning
3. Reinforcement learning.

Describe each category of learning algorithm. Convey difference in the approach adopted by each category.
How well we think each of the following ML algorithms (i) on noisy data (ii) under resourced scenarios.

1. Unsupervised learning.
The model learns from unlabelled data; finding patterns or structures. The algorithm tries to find hidden pattern, similarities or grouping within the data.
Eg:- K-Means clustering
Hierarchical clustering
Association rule mining

Application → Customer segmentation-Market basket analysis pattern discovery.

Approach: The system independently discovers structure and relationship without predefined o/ps.

2. Supervised Learning — the model is trained on labeled data for desired o/p.

→ Maps i/p to get correct o/p.

The algorithm learns the relationship b/w i/p & o/p and predicts results for new unseen data

Application: Email spam detection.
Image classification.
Age prediction

Eg: Linear Regression
Decision tree
Random Forest
Support vector Machine. (SVM).

Approach: The system learns from historical examples with known answers.

3. Reinforcement Learning: is an agent learns by interacting with an environment through trial and error.

Method: — agent receives reward for correct actions
penalties for incorrect actions.

The goal is to maximize cumulative reward over time.

Eg:- Q-Learning, Deep Q networks.

Appl. Use: Robotics, Self driving cars, Game AI

Approach: Systems continuously improves its decision based on feedback from environment

performance on noisy Data

① Supervised learning
→ performance may decrease because incorrect labels affect predictions accuracy.
→ Algorithms like Random forest are more robust to noise.

② Unsupervised learning: Highly sensitive to noisy data because clustering and pattern decision may become inaccurate.

③ Reinforcement learning: Can tolerate some noise because learning occurs over repeated interactions but excessive noise may slow learning.

performance in under resourced Scenarios

means scenarios refer to limited data, low computational power, or limited labelled dataset.

① Supervised learning labelled dataset
→ requires large dataset.
→ performance decrease when labelled dataset is limited.

② Unsupervised learning
Better performed as labelled data is not required

③ Reinforcement learning
→ computational expensive and requires many iterations.
→ usually perform poor in resource constrained environment

Conclusion: B.

What are the common techniques used to detect outliers? What are the benefits and disadvantages of removing outliers from data? How does the presence of outliers impact various ~~of~~ in Statistical Learning.

Ans: Outliers are data points that are significantly differ from rest of the dataset. They may ~~occur due to~~ occur due to

- * Measurement errors
- * Incorrect data entry
- * Genuine rare ~~information~~ observation

Common techniques used to detect outliers.

a) Z-~~Score~~ Score Method
b) IQR (Inter Quartile Range)
c) Box plot method.
d) Distance based and clustering methods

1) Z score

Z-score measures how far a data point is from the mean in terms of standard deviation

$$Z = \frac{X - \mu}{\sigma}$$

where X = data value
μ = mean
σ = standard deviation

~~If z-score > 3 or~~

Rule If z score > 3 or < -3
it is outlier.

Advantages: - Simple and fast
Limitation: Sensitive to non-normal distribution.

2) IQR

The IQR method uses quartiles to detect unusually high or low values.

$$IQR = Q_3 - Q_1$$

Outliers are identified as

Lower bound = $Q_1 - 1.5\,(IQR)$

Upper bound = $Q_3 + 1.5\,(IQR)$.

Advantage:
- Work well for skewed data
- Less effected by extreme values

Limitation
May not work well for small datasets

3) Box plot method

A box plot visually represent the distribution of data and highlight points outside the whiskers as potential outliers.

Advantage: Easy visual interpretation
Limitations: Not ideal for high-dimensional dataset

4) Distance based and clustering method.

Technique such as
(i) K-Means clustering
(ii) DBSCAN &
(iii) Mahalanobis Distance
is used for ML to detect anomaly observation.

Advantage
Effective for multi dimensional data

Limitations: Computational Expensive.

Benefits of Removing Outliers

Removing outliers can improve the quality of ML & Statistical analysis.

Benefits:
- → Improves model accuracy
- → Reduces noise in data
- → Improves normality of distribution
- → prevents skewed statistical measures such as mean and variance
- → Enhances performance of algorithms like Linear Regression & K-means

Drawbacks of removing outliers

- → Important rare events may be removed
- → Loss of valuable information
- → Reduced dataset size
- → May introduce bias in the model.
- → Can reduce model generalization to real world scenarios.

Eg:- Fraud detection often rely on unusual transactions, which may appear as outliers.

Impact of Outliers on Variance in Statistical Learning

Variance $$Var(x) = \frac{1}{n} \sum_{i=1}^{n} (x_i - \mu)^2$$

Outliers greatly increase variance because they are located far from the mean, causing larger squared deviations.

Effects of statistical learning

Increases model instability
Causes overfitting in some algorithms.
Distorts regression line and predictions
Reduces prediction reliability

Conclusion:

You have a classification dataset with 1000 data points. You fit a classifier and get the below confusion matrix

	Actual positive	Actual negative
predicted positive	114686	6
predicted negative	4350	1

Determine Accuracy, precision & recall. and Comment ~~on~~ ~~the~~ ~~model~~ ~~quality~~ of such a model. Discuss one potential method of improving the model.

TP = 114686 TN = 4350

FP = 6 FN = 1

$$\therefore \text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{114686 + 4350}{114686 + 4350 + 6 + 1}$$

$$= 96.34\%.$$

$$\text{precision} = \frac{TP}{TP + FP} = \frac{114686}{114686 + 6} = 99.99\%.$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{114686}{114686 + 4350} = 96.35\%.$$

So overall Acc, precision & Recall is good but still 4350 false values.

this is critical for

- Medical diagnosis
- fraud detection.
- Cyber security

One potential method to improve the model.

1. SMOTE — Synthetic Minority over sampling Technique.
2. undersampling Majority class
3. using class weight.

Benefits: → Reduce false negatives
Improve Recall
Create more balanced learning

Another improvement include:

→ Hyperparameter tuning
→ feature engineering
→ using ensemble methods such as Random Forest
e.g. Logistic

3(a) Curse of Dimensionality and its Impact on clustering algorithms

The curse of dimensionality refers to the problems that arise when the no: of features of dataset becomes very large.

As dimensions increase :
* Data points become sparse
* Distance measures become less meaningful
* Computational complexity increases.

Impact on clustering Algorithms

① Distance Measures become less effective
clustering algorithm such as → K-means
→ Hierarchical clustering
depends heavily on distance calculations.

In high dimensional spaces : → Distance b/w points become very similar
→ difficult to distinguish clusters accurately

② Increased Computational costs:
More dimensions require → more memory.
→ more processing time
This reduces algorithm efficiency.

③ Overfitting Risk. High-dimensional data may contain
→ Irrelevant features
→ noisy attributes

④ Sparse data problem → making pattern discovery difficult

Solutions → Feature selection
→ principal Components analysis.
→ Dimensionality Reduction techniques.

3(b). we classify two groups Yes or No.

Feed Forward Neural N/w & Recurrent Neural N/w.

↓

FFN → is an artificial neural n/w where information moves only in one direction from i/p to o/p.

Main components:

① Input layer
→ Receives i/p data / features.
→ Eg:- Image pixels or sensor values

② Hidden layers: perform computations and feature extraction.
Each neuron weights, bias and activation function

③ o/p layer → produces final prediction.

④ weights and Biases → weights determine importance of inputs.
→ Bias helps shift activation value.

⑤ Activation functions → ReLu, Sigmoid, softmax.

RNN perform better for sequential or time dependent data because they remember previous information.

Applications: → speech recognition
→ Language translation
→ Time series forecasting

Advantages: → Captures temporal dependencies
→ Learn context from previous i/p - data than FFNN
→ Better for sequential

2(a)

Object	Feature 1	Feature 2
O1	1	4
O2	2	1
O3	3	7
O4	4	3
O5	5	6
O6	8	5
O7	9	3
O8	9	7
O9	10	6
10	11	2

So cluster 1 (centroid)

O1 (1, 4)
O2 (2, 1)
O3 (3, 7)
O4 (4, 3)
O5 (5, 6)

cluster 2

O6 (8, 5)
O7 (9, 3)
O8 (9, 7)
O9 (10, 6)
10 (11, 2)

Scikit learn clustering algorithm

$k=2$, Initialize centroid

The algorithm → Initialize centroids
→ Calculate distance
→ Assign point to nearest centroid
→

S(d) = No: of cluster $k=2$,

Initial centroid $C1 = O3 = (3,7)$

$C2 = O9 = (10,6)$

Step 1: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Step 2: Compute new centroid.

new centroid for k1 (1,4), (2,1), (3,7), (4,3), (5,6).

Mean of $\bar{x}$ value = $\bar{x} = \frac{1+2+3+4+5}{5} = 3$

Mean of $\bar{y}$ value = $y = \frac{4+1+7+3+6}{5} = \frac{21}{5} = 4.2$

$\therefore C1' = (3, 4.2)$.

New Centroid for k2 (8,5), (9,3) (9,7), (11,4)

$\bar{x} = \frac{8+9+9+10+11}{5} = 9.5$

$y = \frac{5+3+7+6+2}{5} = 4.6$

CASE STUDY

Problem: Accurately identifying plant diseases early on is crucial for effective treatment and preventing crop loss. Traditional methods rely on expert knowledge which can be time consuming and inconsistent.

Solution: Develop a cloud based image classification model to automatically detect plant diseases from images.

Steps:
① Data collection — Gather a large dataset of plant images with corresponding disease labels.
② Data preprocessing — Resize images to a standard size
→ Augment images (rotation, flipping, cropping) to ~~increase~~ increase dataset diversity.
→ Split data into training, validation and testing sets
③ Cloud platform selection → choose a cloud platform like Aws, GCP or Azure based on project requirement and budget.
④ Model Training → utilize cloud based GPU instances for accelerated training.
→ Experiment with deep learning architectures (Eg:- Convolutional neural, ResNet) and hyperparameters.
→ Train the model on the prepared dataset
⑤ Model Evaluation → Evaluate model performance using metrics like accuracy, precision, recall and F1 score.
→ Iterate on model architecture & hyperparameters to improve performance.
⑥ Model deployment: Deploy the trained model to a cloud based API endpoint.
→ Create a user friendly interface for uploading plant images.
→ Integrate the model with mobile application for on-field use.
⑦ Model monitoring -
→ Continously monitor model performance in production.

Unsupervised ~~Technique for~~ Algorithms for ML

1. K- means clustering
2. DBSCAN clustering (Density Based spacial clustering of application with Noise)

K-means

Centroid based → Assign data points to the nearest centroid, iteratively updating centroid. ie $C = \frac{x_1 + x_2 + \dots x_n}{n}$

No: of clusters → Requires specifying the number of clusters (k) before hand.

Cluster shape → Assumes clusters are spherical.

Sensitivity to outliers: Sensitive to outliers as they can influence Centroid positions.

Efficiency → Generally efficient for large datasets.

DBSCAN Density based Spatial clustering of applications with Noise.

Density based: → Groups together points that are closely packed together

No: of clusters :→ Automatically determines the number of clusters based on data density.

cluster shape → Can discover clusters of arbitary shape.

outliers handling → Explicitly identifies noise points

Efficiency → Can be less efficient for high dimensional data

K-means → suitable for well seperated, spherical clusters with a pre-defined no: of clusters

DBSCAN → Suitable for clusters of arbitary shape, when the no: of clusters is unknown and when dealing with noise and outliers.

retrain the model with new data to adapt to changing conditions.

Non-functional requirements

Scalability — Handle large image datasets and computationally intensive training process.

Cost efficiency — pay only for the resources used, avoiding upfront infrastructure costs.

Accessibility → Collaborate with teams globally through cloud-based platforms.

pre-trained models → Leverage pre-trained models to accelerate development.

Decision Tree

A decision tree is a flowchart like structure used to visualize and predict decisions. It's a supervised ML algorithm that can be used for both classification and regression tasks.

Root node: Represents the entire dataset.

Internal nodes → Represent feature or attributes used to split the data

Branches → Represent the possible outcome of a decision.

Leaf nodes → Represent the final decision or prediction.

Advantages: → Easy to understand & interpret

→ Can handle both numerical & categorical data

→ Requires less data preparation.

Disadvantages

→ Prone to overfitting.

→ Not suitable for all types of data.

